Block Attention for Order-Independent Action Scoring

Independent encoding, a masked architecture, and controlled proofs

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Abstract

When an application presents independent action alternatives, their arbitrary list order should not determine their scores. We distinguish two approaches: a tested workaround that independently encodes evidence and actions using a frozen language model, and Block Attention, a single structured-input architecture that restricts attention between blocks, resets positions within blocks, and uses a shared action–evidence scoring head. We give the associated equivariance argument and report tests on Qwen2.5-0.5B, FLAN-T5-base, and a random 22,144-parameter transformer. The experiments establish a structural property on controlled inputs, not accurate financial decisions. Block Attention is a descriptive name for this construction; block masks and permutation-equivariant attention are established concepts, and novelty is not asserted.

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1 Two methods, two distinct claims

Independent encoding: tested on pretrained models. Application code reads one dictionary, encodes evidence separately, and encodes every action description in a separate model call. It reads hidden vectors rather than generating text. A shared, untrained comparison scores each action against the evidence. Separate files are unnecessary. Existing transformer attention remains active within each input.

Block Attention: tested on a tiny random model. One structured input contains an evidence block and individually delimited action blocks. Application code identifies these spans, supplies a block attention mask, resets positional indices inside each block, pools each action block and applies a shared scoring head. This is not a prompting instruction and is not implemented merely by writing JSON. It requires control over model internals. This action-scoring construction has not been validated as a trained replacement for a pretrained language model's answer generator.

2 Motivation and prior work

Option-order sensitivity has been documented in language-model multiple-choice tasks [1]. Deep Sets [2] and Set Transformer [3] provide established foundations for permutation-invariant and equivariant functions. Our contribution here is an explicit implementation and controlled demonstration of action isolation for a financial workflow. The behavior does not identify a defective attention head or isolate attention from positional encoding and learned behavior as the cause of a pretrained model's failure. Sequence packing already combines attention isolation and adjusted positional embeddings to preserve separate-sequence computation [4]. UniLM combines different attention masks in a shared transformer [5]; Fusion-in-Decoder independently encodes passages and combines them through decoder cross-attention [6]. Thus neither block isolation nor mixed attention patterns are new in themselves.

3 The mathematical construction

Let $X = (E, A_1, \ldots, A_m)$ denote token blocks, where E is evidence and A_a contains an action name and definition. Each token has block ID b_i and within-block position r_i. For a causal decoder, define

$$M_{ij} = \begin{cases} 0, & b_i = b_j \text{ and } r_j \le r_i, \\ -\infty, & \text{otherwise.} \end{cases} \tag{1}$$

For each transformer attention head,

$$\mathrm{BA}(X) = \mathrm{softmax}_{\mathrm{row}}\left(\frac{QK^\top}{\sqrt{d_k}} + M\right)V. \tag{2}$$

The mask forbids cross-block attention. It does *not* suffice alone: token representations must use local positions r_i, not their global positions in the concatenated prompt. With learned positional embeddings, initialize $h_i^{(0)} = e(t_i) + p(r_i)$. With rotary or other position mechanisms, their indices and any relative biases must respect block-local coordinates too. The mask must hold in every layer; residuals, feed-forward networks and normalization must not mix blocks.

Let F_θ be the frozen transformer. Normalize evidence-token rows to obtain $H \in \mathbb{R}^{n_E \times d}$. Independently pool and normalize each action block:

$$q_a = \mathrm{norm}\left(\frac{1}{|A_a|}\sum_{i \in A_a} h_i^{(L)}\right). \tag{3}$$

The exact untrained head used in our tests is

$$w_a = \mathrm{softmax}(Hq_a), \quad z_a = H^\top w_a, \quad s_a = q_a^\top z_a, \quad c_a = \frac{\exp s_a}{\sum_b \exp s_b}. \tag{4}$$

There are no additional learned projections or fitted temperatures. c_a is a normalized compatibility score, not a language-generation probability or calibrated belief. Evidence and action blocks first remain isolated; the explicit head subsequently allows action–evidence interaction.

Proposition 1. *Assume fixed tokenization within blocks, shared weights, local positions, the mask above at every layer, tokenwise residual/normalization/feed-forward operations, and deterministic inference. Permuting action blocks permutes their score rows: $S(E, PA) = PS(E, A)$. Named-action scores are invariant in exact arithmetic.*

Proof. Local positions make each block's initial representations independent of its concatenation position. The mask restricts every attention sum to the same within-block tokens. By induction over layers, each block has the same hidden states under reordering. Shared pooling and action–evidence comparison therefore preserve the same named-action score. The denominator is symmetric in the action scores. □

In software, normalize in canonical action-name order and use an order-independent tie-break. Use tolerances for floating-point changes. Ordinary autoregressive answer generation is not covered by the proposition; an output token attending across blocks can reintroduce order sensitivity. Evidence-record order and action paraphrases are also outside the guarantee.

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4 Pretrained-model workaround: measured results

Preliminary results: frozen pretrained models

Five constructed EPS cases cover above/below/equal consensus, missing actual EPS and conflicting actual EPS. Actions are BUY, CASH and WITHHOLD, not BUY/HOLD/SELL. Each case has six action orders and five identical repeats: 150 evaluations per method. There are 75 within-case pairs of distinct orders and 300 within-case/order pairs of repeats. These are five independent input scenarios, not 150 independent financial observations.

The prompt baseline serializes the dictionary into one input. All three candidate answers are teacher-forced and their token log probabilities summed, including EOS. Scores are normalized over those three complete answers. Qwen was also scored without EOS as a control. No independent generated-answer evaluation was performed in this experiment. The alternative uses independently encoded representations and Equation (4).

Model	Measure	JSON prompt	Separate encoding
Qwen 0.5B	Mean order TV	0.2505	0
	Maximum order TV	0.5975	0
	Changed top rank / 75	32	0
	Maximum repeat TV	0	0
FLAN-T5-base	Mean order TV	0.0546	0
	Maximum order TV	0.1488	0
	Changed top rank / 75	0	0
	Maximum repeat TV	0	0

Here $\mathrm{TV}(p, q) = \frac{1}{2}\sum_a |p_a - q_a|$. Qwen used 494,032,768 parameters and 111.93 seconds on CPU; FLAN used 247,577,856 parameters and 51.64 seconds. Weights were frozen. CPU float32 inference was deterministic. Repeated inputs were recomputed; invariant results were not returned from an answer cache. In Qwen, removing EOS still gave 32 changed rankings and maximum TV 0.5969.

5 Direct Block Attention proof: tiny-model ablations

Preliminary results: masked tiny transformer

A two-layer transformer with width 32, four attention heads, tokenwise layer normalization and a 64-unit feed-forward hidden layer has 22,144 parameters. All weights are random and fixed (seed 20261020). The same five-case/six-order/five-repeat design compares three conditions with identical weights and the same pooling/scoring head.

Condition	Max. order TV	Changed ranks / 75	Max. forbidden attention
Ordinary causal attention	0.04117	52	0.96843
Block mask, global positions	0.04879	51	0
Block mask, local positions	0	0	0

All repeat distances were zero. In the full Block Attention condition, changing an earlier BUY description, including its length, changed the CASH vector by exactly zero. All forbidden attention entries in both layers were zero. Tests asserted invariance within 10^{-6}; measured differences were zero. The mask-only ablation demonstrates why global positions must also be removed. This is an architectural test on one random initialization, not a trained small language model or pretrained-model retrofit.

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6 Interpretation, implementation and limits

The practical workaround bypasses the vocabulary output head and decoding, not attention within the model. Block Attention changes the attention connectivity of a single structured input. Both methods isolate action representations before a shared evidence comparison. The tiny-model proof does not establish that a mask retrofit preserves useful pretrained representations.

The stipulated EPS policy was matched on 60/150 Qwen runs in each method and 30/150 FLAN runs in each method. We did not optimize decision accuracy, train a scoring head, test returns, or use independent investment-outcome labels. The purpose is solely to establish action-order behavior. The baseline probabilities and alternative compatibility scores have different meanings; their absolute levels are not interchangeable. Independent repeats rule out observed run-to-run variation on this setup, not population-wide instability or hardware effects.

A production implementation must specify trusted parsing of action boundaries, position handling, padding and cache isolation, the scope of normalization, a shared head, an order-independent tie policy, and numerical tolerances. Learned cross-action attention could be included if it is permutation equivariant; isolation is a sufficient construction, not the only solution. Dependent action plans should not be treated as an unordered choice set.

7 Reproduction artifacts

- `dictionary_invariance/run.py`: FLAN independent-encoding comparison.
- `dictionary_invariance/qwen/run.py`: Qwen comparison with EOS and prefix scores.
- `block_attention/run.py`: tiny model, masks, positional ablation and isolation test.
- Each directory includes machine-readable outputs and `summary.json`.

Model revisions:

Qwen: `7ae557604adf67be50417f59c2c2f167def9a775`.

FLAN: `7bcac572ce56db69c1ea7c8af255c5d7c9672fc2`.

The recorded environment uses PyTorch 2.13.0 and Transformers 4.57.3 for pretrained models. Tiny-model execution takes approximately 0.15 seconds after imports on this CPU; it is not a training-time estimate.

References

[1] P. Pezeshkpour and E. Hruschka (2024). Large Language Models Sensitivity to the Order of Options in Multiple-Choice Questions. Findings of NAACL. https://aclanthology.org/2024.findings-naacl.130/

[2] M. Zaheer et al. (2017). Deep Sets. NeurIPS. https://arxiv.org/abs/1703.06114

[3] J. Lee et al. (2019). Set Transformer: A Framework for Attention-based Permutation-Invariant Neural Networks. ICML. https://proceedings.mlr.press/v97/lee19d.html

[4] M. M. Krell et al. (2021/2022). Efficient Sequence Packing without Cross-contamination: Accelerating Large Language Models without Impacting Performance. https://arxiv.org/abs/2107.02027

[5] L. Dong et al. (2019). Unified Language Model Pre-training for Natural Language Understanding and Generation. https://arxiv.org/abs/1905.03197

[6] G. Izacard and E. Grave (2021). Leveraging Passage Retrieval with Generative Models for Open Domain Question Answering. EACL. https://aclanthology.org/2021.eacl-main.74/

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Figure 1: the tested independent-encoding workaround

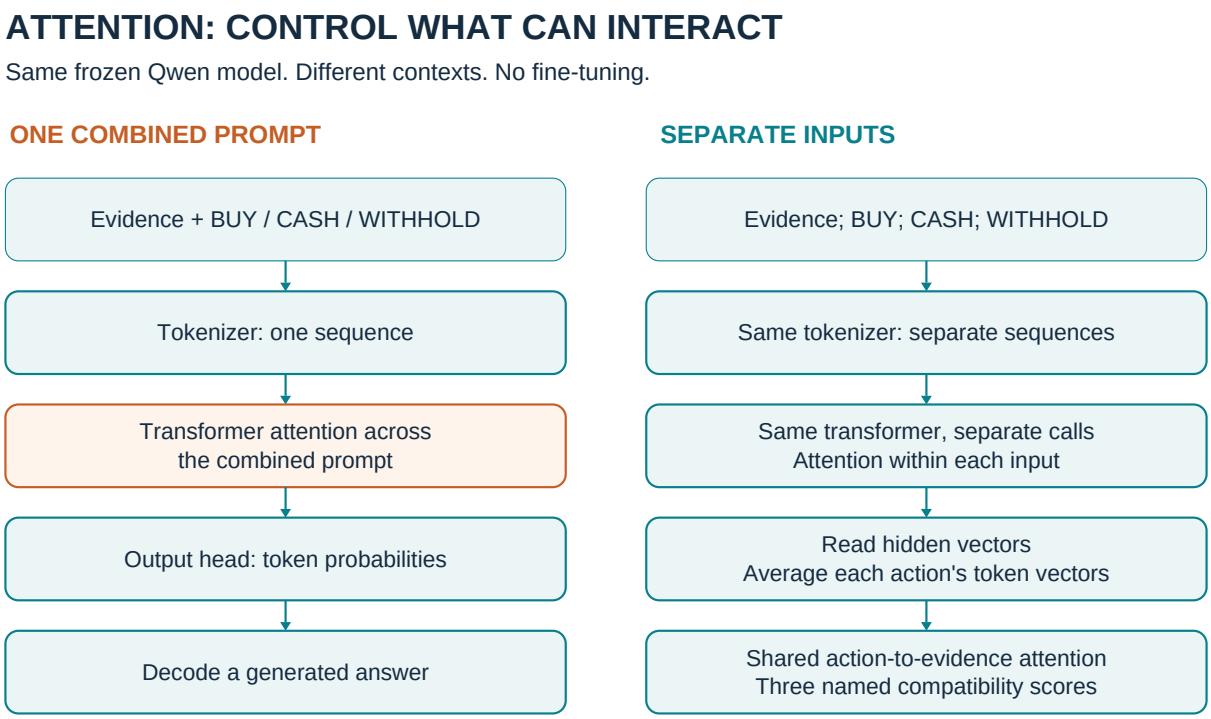
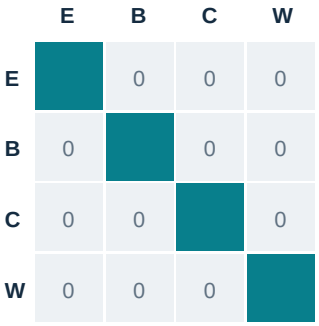


Figure 2: Block Attention and its positional ablation

BLOCK ATTENTION

One structured input. Isolated attention blocks. Local positions.



1. Parse evidence and action spans.

2. Block attention between spans.

3. Restart positions inside each span.

4. Pool actions; share the scoring head.

Attention remains active within every block.

E: evidence | B: BUY | C: CASH | W: WITHHOLD
Teal: within-block causal attention; grey: blocked.

PRELIMINARY ARCHITECTURAL TEST

22,144 random parameters | 2 layers | 4 heads | No training

Same model and inputs	Changed ranks / 75	Maximum TV
Ordinary attention	52	0.04117
Block mask only; global positions	51	0.04879
Block mask + local positions	**0**	**0**

Forbidden attention weights: exactly zero. Repeated-run variation: zero.

Tested on a tiny random model, not a masked retrofit of pretrained Qwen.

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